

WISN Core: Beginner's Project Walkthrough

A two-day teaching guide, developer walkthrough, and evaluator Q&A

Audience: A first-time presenter with little or no Laravel experience
Purpose: Teach what the project does, how its code and data fit together, what technologies it uses, and how to answer likely academic questions.
Repository basis: [niranjanrimal25/wisn-core](#), reviewed on 7 October 2026 against the current Operations redesign and architecture notes.

Use this guide honestly. It describes the code that is in this repository. It separates the annual WISN planning calculation from the newer, manual shift-operations prototype. It does not claim that the application is connected to a hospital EHR, contains real patient records, or automatically assigns nurses.

1. How to use this guide

The learner does not need to memorize Laravel syntax. By the end, they should be able to explain, in ordinary language:

1. What problem WISN Core addresses and what WISN means.
2. What PHP, Laravel, a database, Blade, Tailwind, Vite, Chart.js, and DomPDF each do.
3. How information travels from a form to the database and back to a page.
4. How AWT, workload standards, staff requirements, and the WISN ratio are calculated in this code.
5. What the operations module adds—and what it still does not automate.
6. What evidence supports the demo data, and which limitations must be acknowledged.

Teaching rule: after each section, ask the learner to explain it back without reading. Correct the idea first; only then show the relevant code file.

2. A two-day crash-course plan

Day 1 — Understand the project and trace the data

Time	Lesson	Teaching outcome
09:00–09:30	Project story and 60-second introduction	Learner can say what the application is for.
09:30–10:30	WISN terms and one hand calculation	Learner understands AWT, time standard, annual volume, FTE, and ratio.
10:45–11:30	Technology stack using simple analogies	Learner can name each tool and its job.
11:30–12:15	Laravel MVC and request flow	Learner can explain route → controller → service/model → view.

Time	Lesson	Teaching outcome
13:00-14:00	Database tables and relationships	Learner can describe departments and workload activities.
14:00-15:00	File-by-file guided tour	Learner can find the code that implements each feature.
15:15-16:00	Dashboard, charts, and PDF report	Learner can explain how results are displayed.

Day 2 — Explain the implementation, limits, and defend it

Time	Lesson	Teaching outcome
09:00-10:00	Walk the WISN calculator line by line	Learner can trace inputs into the returned result.
10:00-10:45	Operational staffing overlay	Learner can distinguish a shift snapshot from annual WISN.
11:00-11:45	Data provenance, privacy, and security	Learner avoids unsupported claims about real-world data or access control.
12:30-13:15	Run-through of a short live demo	Learner can demonstrate the application in a clear order.
13:15-14:15	Professor/evaluator Q&A	Learner practices short, accurate answers.
14:30-15:00	Final mock presentation	Learner answers follow-up questions without relying on the guide.

Minimum rehearsal test: the learner should be able to explain the project in 60 seconds, trace one dashboard request in 60 seconds, and do the AWT example below without assistance.

3. The project in one minute

Plain-language explanation

WISN Core is a browser-based nursing workforce-planning prototype for a hospital. A manager enters each department's available working-time components and the annual volume and time standard for workload activities. The application calculates an estimated staffing requirement and WISN ratio, then shows the results in a dashboard and downloadable PDF report. A separate Operations page records observed inpatient census at 12:00 AM, 7:00 AM, and 7:00 PM, plus actual nurses on duty at the two handovers. It calculates required nurses only from an effective, locally approved target and suggests aggregate moves for human review. It is not connected to an EHR or roster and does not assign individual nurses.

Problem it addresses

Manual staffing calculations can be time-consuming, hard to repeat, and difficult for managers to interpret. The project digitizes the selected WISN calculation steps, validates common inputs, and displays a consistent department-by-department summary.

Main users

- Hospital or facility administrators reviewing overall staffing.
- Nurse managers entering department workload and staffing information.
- Academic demonstrators showing how a health-workforce method can be implemented as a web application.

What it is—and is not

It is	It is not
A Laravel web application with a server-rendered interface.	A mobile app, microservice system, or React/Vue single-page app.
A prototype that calculates staffing estimates from entered data.	A replacement for clinical judgment or workforce-policy approval.
A single-facility data model in the current repository.	A multi-hospital system with tenant isolation.
An annual/reference-period WISN workflow plus a manual operational overlay.	A live feed from patient records, staff attendance, or a roster system.
A tool that displays aggregate counts and reports.	A system that currently assigns named nurses or executes transfers.

4. WISN explained for a beginner

WISN means **Workload Indicators of Staffing Need**. It is a World Health Organization approach to estimating how many health workers are needed for a defined workload. In this project, the cadre is nursing staff and the unit of analysis is a hospital department.

The high-level question is:

Given the work a department performs, the time each task takes, and the time one nurse is available to work in a year, how many nurses are needed?

Key terms

Term	Beginner-friendly meaning
Available Working Time (AWT)	Productive hours one nurse is available to work over the chosen year, after subtracting non-working time.
Workload activity	A task included in the calculation, such as triage, medication administration, or teaching.
Time standard	The expected hours needed for one occurrence of an activity—or, for support tasks, the hours used in a shift.
Annual volume	How many times a health-service or additional activity occurs during the year.
Standard workload	The number of occurrences one nurse could perform in a year at the entered time standard.

Term	Beginner-friendly meaning
FTE	Full-time equivalent. A value such as 1.5 means workload equivalent to one and a half full-time staff, not necessarily a specific roster.
WISN ratio	Current department staff divided by estimated required staff.
Support allowance / CAF	An adjustment for time spent on recurring support activities.
Additional activity / AAF	In this application's formula, a separate FTE contribution calculated from its annual hours.

Important time-scale distinction

The main WISN calculation uses annual workload volumes and annual available working time. It can be recalculated when a user updates those inputs, but that does not make it an intraday staffing allocator. It answers a planning question over a reference period; it does not know who is on duty right now unless operational data is separately supplied.

5. The technology stack

The stack is visible mainly in `composer.json`, `package.json`, `config/`, and the `resources/` directory.

Tool	What it is	What it does here
PHP 8.1+	Server-side programming language.	Runs Laravel, controllers, models, and the calculation service. <code>composer.json</code> declares PHP ^8.1.
Laravel 10	PHP web framework.	Provides routes, middleware, validation, database access, authentication scaffolding, views, and application structure.
Composer	PHP package manager.	Installs Laravel and PHP libraries declared in <code>composer.json</code> / <code>composer.lock</code> .
MySQL (default example)	Relational database.	Stores users, departments, activities, and operational snapshots. The actual connection is set in <code>.env</code> ; Laravel's config also supports SQLite and other drivers.
Eloquent ORM	Laravel's object-relational mapper.	Lets code work with <code>Department</code> and <code>WorkloadActivity</code> models rather than writing routine SQL by hand.
Laravel migrations	Versioned database schema scripts.	Create and evolve tables in a repeatable way.
Blade	Laravel's server-side template engine.	Combines application data with HTML templates under <code>resources/views/</code> .

Tool	What it is	What it does here
Tailwind CSS 3	Utility-class CSS framework.	Styles the pages using classes such as spacing, color, layout, and responsive breakpoints.
Vite 4	Front-end development/build tool.	Loads and builds the CSS and JavaScript assets referenced by the Laravel layout.
Alpine.js 3	Small JavaScript interaction library.	Supports lightweight page interactions such as menus and toggles.
Chart.js 4.4	JavaScript charting library.	Draws dashboard charts; the dashboard loads it from a CDN.
DomPDF	PHP PDF renderer (barryvdh/laravel-dompdf).	Converts a Blade report template into a downloadable PDF.
Laravel Breeze	Authentication starter kit.	Supplies login, registration, password-reset, and profile scaffolding.
PHPUnit 10	PHP testing framework.	Runs unit and feature tests. The repository's test environment is configured to use in-memory SQLite.
Git / GitHub	Version-control tools.	Track the source and its development history.

Composer versus npm

- **Composer** handles PHP/Laravel packages.
- **npm** handles browser-side development packages such as Tailwind, Alpine.js, and Vite.
- Installing one does not install the other.

What not to overstate

`composer.json` also declares packages such as Sanctum and Guzzle, but the primary browser workflow is the authenticated Laravel `web.php` flow. Do not describe the product as API-first just because an API route file or package exists.

6. Architecture using a simple analogy

Think of a small hospital office:

Software part	Office analogy	Project examples
Route	Receptionist: decides where a request should go.	<code>routes/web.php</code> , <code>routes/auth.php</code>
Middleware	Door check: verifies the request is allowed to continue.	<code>auth</code> , <code>verified</code> route middleware

Software part	Office analogy	Project examples
Controller	Coordinator: validates a request and asks the right part of the system to do work.	<code>DashboardController</code> , <code>DepartmentController</code>
Model / ORM	Organized record book: represents and retrieves stored data.	<code>Department</code> , <code>WorkloadActivity</code>
Service	Specialist calculator: holds business rules and calculations.	<code>WisdCalculatorService</code> , <code>OperationalStaffingService</code>
Database	Filing cabinet: persists records after a page closes.	MySQL configured by <code>.env</code>
View	Printed/displayed form: turns prepared data into a page.	<code>resources/views/dashboard.blade.php</code>
Migration	Database construction plan: creates a table consistently.	Files under <code>database/migrations/</code>

Request path diagram

```

Browser
-> routes/web.php
-> auth / verified middleware
-> controller
-> model and/or calculation service
-> database query or update
-> Blade view (HTML) or DomPDF (PDF)
-> response back to browser

```

This is a **Laravel monolith**: the backend, server-rendered pages, and business logic live in one application. The major user interface is not a separate frontend application.

Why put calculation logic in a service?

`WisdCalculatorService` keeps the WISN maths out of the page controller. The dashboard and PDF report can use the same calculation logic. It is easier to inspect and test one calculation service than to duplicate the formula in multiple views or controllers.

7. Database: what is stored and how it relates

A relational database stores related facts in tables. The key relationship is:

```
Department 1 ---- many WorkloadActivity rows
```

One department can have many workload activities; each activity belongs to one department.

Core tables

Table	What it stores	Important fields / links
<code>users</code>	Application login accounts.	Name, email, password hash, verification timestamp, and the <code>is_admin</code> flag used for ratio configuration access.

Table	What it stores	Important fields / links
<code>departments</code>	One row per hospital department.	Name, type, <code>current_staff</code> , annual workdays, holidays, leave, training, and hours per day.
<code>workload_activities</code>	One row per task included in WISN.	<code>department_id</code> , name, type, time standard, annual volume. Department deletion cascades to these activities in the original schema.

Laravel also has framework tables for password-reset tokens, failed jobs, and Sanctum personal access tokens. Those are not the WISN calculation inputs.

Department fields and AWT

The department stores the components used to calculate AWT:

- `working_days_per_year`
- `public_holidays`
- `annual_leave_days`
- `sick_leave_days`
- `training_days`
- `working_hours_per_day`

AWT itself is computed by an Eloquent accessor; it is not stored as a separate current column. An earlier migration created `available_working_time_hours`, and a later migration removed it after the breakdown fields became the source of truth.

Activity types

Stored value	Purpose	Annual volume
<code>health_service</code>	Direct service activity, such as a clinical task.	Required by controller validation.
<code>support</code>	Recurring shift work, such as handover.	Not used by the code for the support fraction.
<code>additional</code>	Additional activity such as teaching.	Required by controller validation.

Operational tables added alongside WISN

Table	What it stores
<code>operational_staffing_standards</code>	Admin-configured patient-per-nurse target and optional minimum by inpatient unit type and day/night handover, with source, version, effective dates, and approval metadata. No ratio is preloaded.
<code>operational_rounds</code>	Daily census date, 12:00 AM / 7:00 AM / 7:00 PM handover code, scheduled observation time, actual capture time, entering user, and optional note. Corrections create a newer version.

Table	What it stores
<code>department_operational_snapshots</code>	Directly observed unit patient count; actual nurses on duty at 7 AM / 7 PM only; applicable standard and calculated required nurse count when configured.
<code>mobilization_recommendations</code>	Aggregate count from a sending unit to a receiving unit; status, rationale, and reviewer/time fields.

The operations module uses separate census and `on_duty_staff` snapshot values. It does **not** overwrite `departments.current_staff`, which is the WISN department headcount. OPD is excluded; Emergency counts refer to admitted or observation-bed patients, not walk-in visits.

Who owns the department rows?

In this prototype, there is no `hospital_id` and no user-to-department ownership link. Departments are shared across accounts. This is a single-facility data model, not multi-tenant access control.

8. WISN calculation: step by step

The implementation is in `app/Services/WisnCalculatorService.php`. Its main entry point is:

```
calculateDepartmentStaffing(Department $department): array
```

It returns an array containing the department details, AWT, health-service FTE, support multiplier, AAF FTE, total required staff, WISN ratio, status, and an activity breakdown. It does not write the calculated result back to a database table; the controller passes the returned array to a view.

Step 1 — Calculate AWT

```
AWT = (working days - public holidays - annual leave - sick leave - training) × hours/day
```

Using the sample/default values in the code:

```
(260 - 13 - 18 - 12 - 5) × 8
= 212 × 8
= 1,696 hours per nurse per year
```

Step 2 — Health-service standard workload and staff

For each `health_service` activity:

```
Standard workload = AWT / time standard
Required FTE for activity = annual volume / standard workload
```

A larger annual volume or longer time per task increases required FTE. A higher AWT increases the amount one nurse can contribute over the year.

Step 3 — Support activities and CAF

For each `support` activity, the code calculates its fraction of one working day:

```
Support fraction = support hours per shift / working hours per day
Total support fraction = sum of support fractions
CAF = 1 / (1 - total support fraction)
```

The code uses the department's working-hours-per-day value; it does not hard-code an eight-hour shift. The multiplier is used to adjust the health-service FTE.

Step 4 — Additional activities and AAF

The current code uses this project-specific formula:

```
Annual additional hours = sum(annual volume × time standard)
AAF FTE = annual additional hours / AWT
```

The AAF FTE is added to the health-service FTE after the CAF adjustment.

Step 5 — Total and ratio

```
Total required staff = (health-service FTE × CAF) + AAF FTE
WISN ratio = current department staff / total required staff
```

Interpretation:

- Ratio below 1.00: current headcount is below the model's estimated requirement.
- Ratio equal to 1.00: current headcount equals the estimate.
- Ratio above 1.00: current headcount is above the estimate.

The status labels in this application are: `no_data` for a non-positive ratio, `critical` below 0.90, `borderline` from 0.90 up to but not including 1.00, `adequate` at exactly 1.00, and `surplus` above 1.00.

Worked example using this code's formulas

This is a teaching example, not a hospital recommendation:

1. AWT is 1,696 hours/year.
2. One health-service task takes 0.5 hours; annual volume is 3,392 tasks.
3. Standard workload is $1,696 / 0.5 = 3,392$ tasks per nurse per year.
4. Health-service requirement is $3,392 / 3,392 = 1.00$ FTE.
5. One support activity takes 0.5 hours in an 8-hour day: support fraction $0.5 / 8 = 0.0625$ (6.25%). CAF is $1 / (1 - 0.0625) = 1.0667$.
6. An additional activity occurs 1,696 times and takes 0.5 hours each: $848 \text{ hours} / 1,696 \text{ AWT} = 0.50$ AAF FTE.
7. Total required is $(1.00 \times 1.0667) + 0.50 = 1.5667$ FTE (displayed to two decimals as 1.57).
8. If current staff is 2, the ratio is approximately $2 / 1.5667 = 1.28$, so the code labels the department `surplus`.

Formula caveat to communicate

The guide above describes the **formula currently coded in this repository**. It should not be presented as proof that every WISN methodological assumption has been independently certified. In particular, clinical time standards, workload volumes, support allowances, additional-activity treatment, and category definitions should be checked with the relevant WISN/health-workforce experts before operational use. The current service applies CAF only when the support fraction is between 0 and 1; invalid allowance inputs should be validated and reviewed.

9. Walkthrough of the main user journeys

A. Dashboard: `GET /dashboard`

1. `routes/web.php` maps `/dashboard` to `DashboardController@index`.
2. The route is inside the authentication middleware group.
3. `DashboardController` queries departments and eager-loads their activities.
4. It calls `WISNCalculatorService` once for each department.

5. It adds up current and required staff to calculate a facility-level ratio: `sum(current) / sum(required)`. This is a weighted facility ratio, not the average of department ratios.
6. It passes the result arrays to `resources/views/dashboard.blade.php`.
7. Blade renders the page, tables, cards, and Chart.js charts.

Files to show: `routes/web.php`, `app/Http/Controllers/DashboardController.php`, `app/Services/WisnCalculatorService.php`, `resources/views/dashboard.blade.php`.

B. Department management

1. The user opens a department form.
2. The browser sends a POST or PUT request.
3. `DepartmentController` validates the fields and rejects a non-positive net working-day count.
4. Eloquent writes the department row.
5. The browser is redirected to the department list.

Files to show: `app/Http/Controllers/DepartmentController.php`, `app/Models/Department.php`, `resources/views/departments/`, department migrations.

C. Workload activities

1. Activity routes are nested under a department, for example `/departments/{department}/activities`.
2. The controller validates name, type, time standard, and annual volume rules.
3. The activity is stored with a `department_id` foreign key.
4. The next dashboard/report request loads the updated activities and recalculates the WISN result; calculations are not cached in a separate results table.

Files to show: `app/Http/Controllers/WorkloadActivityController.php`, `app/Models/WorkloadActivity.php`, `resources/views/activities/`, workload-activity migration.

D. PDF report: GET /report

1. `ReportController` loads departments with activities.
2. It calls the same WISN service.
3. It prepares an AWT breakdown and facility totals.
4. `Pdf::loadView()` renders `resources/views/reports/wisn-summary.blade.php` through DomPDF.
5. The browser downloads the report.

The PDF view uses inline CSS because it is rendered by the server-side PDF engine rather than the browser's normal asset pipeline.

E. Operations census: POST /operations/snapshots

1. A user selects a date and one fixed observation time: 12:00 AM, 7:00 AM, or 7:00 PM.
2. The controller requires a directly observed patient count for every inpatient nursing unit; it also requires actual on-duty nurses at the 7 AM and 7 PM handovers. OPD is excluded.
3. `OperationalStaffingService` writes a timestamped round and unit snapshots in a database transaction.
4. For day/night only, it looks up the approved, effective standard for each unit type and shift. It calculates required nurses only when a standard exists; otherwise the requirement is left blank.
5. It marks prior pending proposals as superseded and suggests aggregate count moves only between units with configured requirements.
6. The page shows each handover and daily unit timeline. A human must review any potential move.

Census is entered as observed at each handover; the application does not derive it from unrecorded admissions, discharges, or transfers.

Files to show: `OperationalSnapshotController.php`, `OperationalStaffingService.php`, `OperationalStaffingStandardController.php`, operational migrations, `resources/views/operations/index.blade.php`, and `resources/views/operations/standards.blade.php`.

10. The operational census and shift-staffing view: current behavior

The Operations feature keeps a separate, shorter time horizon beside WISN:

- **Annual planning:** WISN uses annual workload and available working time to estimate a department's longer-term staffing requirement.
- **Midnight census:** enter the observed patient count at exactly 12:00 AM. This is a point-in-time census only; no on-duty nurse count or ratio is applied.
- **Day handover:** at 7:00 AM, enter observed patients and the number of nurses who actually came on duty.
- **Night handover:** at 7:00 PM, enter observed patients and the number of nurses who actually came on duty.

Each handover census is entered directly. The application does not derive 7 AM or 7 PM counts from admissions, discharges, or transfers. The selected date runs from 12:00 AM to 11:59 PM, and the date picker lets the user review prior days. Corrections are saved as a new version so the prior record remains in history.

Configurable targets—not a universal ratio

An administrator may configure a database-backed target by inpatient unit type and day/night shift. A target records patients per nurse, any minimum nurses per shift, the source and version, effective dates, and who activated/approved it. No ratios are preloaded. A requirement is calculated only if the target is approved and effective for that unit, shift, and date:

Required nurses = max(approved minimum, ceiling(observed patients ÷ patients per nurse))

If no suitable target exists, Operations still records the census and nurses on duty, but leaves required nurses blank and makes no move suggestion for that unit. The user must verify and approve the facility's chosen source before activating a target.

Cross-unit balance and human review

When both donor and receiving units have configured targets, the service compares actual on-duty nurses with calculated requirements. For example, using **hypothetical approved demo standards only**:

Unit	Observed patients	Nurses on duty	Calculated required	Balance
Medical Ward	4	3	1	+2
Surgical Ward	9	1	3	-2

The service may suggest **2 aggregate nurses from Medical Ward to Surgical Ward**. That is a potential move for qualified human review, not an assignment or instruction.

Safety and scope boundaries

- Operations captures patient census and actual on-duty nurses separately from annual WISN headcount and annual turnover. It does not update `departments.current_staff` or annual WISN results.
- OPD is excluded. Emergency counts must include admitted or observation-bed patients, not walk-in visits.
- No named nurse roster, skill mix, patient identity, or EHR integration exists. Do not enter patient names or identifiers in notes.
- The application does not check competency, specialty, breaks, shift limits, patient acuity, legal rules, or whether a move is safe. A qualified person must assess these before acting.
- Approval records a human decision; it does not execute a transfer, change actual counts, or modify a roster.

- New snapshots supersede unreviewed recommendations. Review is blocked if the data is older than the configured 30-minute default, the unit set changed, or the move would breach a calculated source/destination target.
- Only an administrator can configure or activate standards. Authenticated users can enter census and review aggregate proposals; `verified` middleware is listed on the routes, but the current `User` model does not enforce email verification.

How to describe the feature in a presentation

Say: **“Operations is a separate manual inpatient census and shift-staffing view. It records observed patient counts at midnight and at the 7 AM and 7 PM handovers, plus actual on-duty nurses at the two staffed handovers. Required counts come only from an effective, locally approved unit-and-shift target. Potential moves are aggregate and require human review; annual WISN remains separate.”**

Do not say: “The system is connected to live patients,” “the app calculates a ratio that has not been approved,” or “approval automatically sends a nurse to another unit.” Those statements are not supported by the code.

11. Where the information came from

Question	Where to look in this repository	What it tells you
What is the intended academic problem and scope?	<code>WISN_Core_Project_Report.md</code> , <code>WISN_Proposal_Final (1).pdf</code>	Background, aims, scope, and project rationale.
What is the current software architecture?	<code>ARCHITECTURE.md</code>	Routes, models, calculation service, views, and current operational overlay.
What does the code actually calculate?	<code>app/Services/WisnCalculatorService.php</code>	Exact formulas, rounding, status thresholds, and guards.
What are the table fields and relationships?	<code>database/migrations/</code> , <code>app/Models/</code>	Database structure and Eloquent relationships.
What sample data appears in the app?	<code>database/seeder/DatabaseSeeder.php</code> <code>NepalDataSeeder.php</code>	Demo departments, activities, and comments about data provenance.
Which PHP libraries are declared?	<code>composer.json</code> , <code>composer.lock</code>	Framework and PHP dependencies.
Which front-end tools are declared?	<code>package.json</code> , <code>vite.config.js</code> , <code>resources/js/</code> , <code>resources/css/</code>	Vite, Tailwind, Alpine, and browser assets.
How is the browser flow routed?	<code>routes/web.php</code> , <code>routes/auth.php</code>	Page endpoints and authentication scaffold routes.
What is the original WHO reference?	<code>9789240070066-eng.pdf</code> and the reference list in the project report	Methodology reference material included in the repository.
What Nepal context material is included?	<code>Nepal Health Facility Data Summary.pdf</code> , <code>WISN_Core_Source_Access_Sheet.pdf</code>	Context/source documents bundled with the project.

Data provenance: an important answer

The repository contains reference PDFs and seeders whose comments describe Nepal-oriented demo assumptions and sources. The running application does not automatically query those PDFs or connect to a government database. Seeded data are sample/demo values, not proof that the application contains actual hospital records or that every entered standard is an official national standard. For an academic presentation, cite the exact source for any specific number and label simulated data clearly.

Annual turnover note

The wider project may include a separately calculated annual nurse-turnover result. In the Laravel source inspected for this guide, there is no turnover table, model, controller, or calculator service wired into the app. Present turnover as a separate result/module unless the presenter can show its actual integration point. It is a longer-term workforce-supply signal; it is not an input to the current WISN formula or the operations page.

12. Start-to-finish teaching walkthrough: exactly what to open and say

Use this as an **18-22 minute guided code tour**. It is written as presenter instructions, not just a list of files. Keep the repository open in an editor and, if the application runs on the presentation machine, keep the browser open too. On Windows/Linux use **Ctrl+P** to open a path and **Ctrl+F** to find a symbol; on Mac use **Cmd+P** and **Cmd+F**. Do not read whole files aloud: find the named class, method, route, or phrase, point to the few relevant lines, explain them, then follow the transition to the next file.

Before the walkthrough (2 minutes)

1. Open the repository root, `niranjanrimal25/wisn-core`, in the editor.
2. Keep `ARCHITECTURE.md`, the application dashboard (if runnable), and this guide available.
3. Use only a disposable demo database if you will show seeded data. The current main seeder deletes/replaces department and activity demo rows; do not seed a database with real operational data.
4. If the local app cannot run, do not pretend it did. You can still teach the request path using the source and the screenshots/report available to you.

Opening words: “I’ll trace one annual WISN result from the browser route, through the PHP calculation, to the page and PDF. After that I’ll show the separate shift snapshot feature, because it answers a different question.”

Part A — trace the annual WISN feature

Step 1 — Show the project map (about 1 minute)

Open: `ARCHITECTURE.md` at the repository root. Do not use the root `README.md` as the project overview; it is still the stock Laravel starter README.

Point to: the “Request flow” and “Service layer” sections. Briefly identify the route, controller, model, service, database, and Blade view. Explain that `ARCHITECTURE.md` is a map; the source files that follow are the evidence.

Tell them: “This is one Laravel application. A browser request comes in through a route, a controller coordinates the work, models read the stored records, a service performs the WISN maths, and a Blade view displays the result.”

Move on: “First, let’s identify the tools this application uses.”

Step 2 — Identify the tools (about 1 minute)

Open: `composer.json`, then `package.json`.

Find/show: In `composer.json`, point to the PHP requirement and the Laravel and `barryvdh/laravel-dompdf` dependencies. In `package.json`, point to the Vite build scripts and the Tailwind/Alpine dependencies. Chart.js is loaded from a CDN in the dashboard view, so do not say it is installed by `package.json`.

Tell them: “Composer installs the PHP/Laravel side. npm installs and builds the browser-side CSS and JavaScript. They are two separate package managers. Laravel and Blade render the page; Chart.js draws the dashboard charts; DomPDF creates the downloadable report.”

Move on: “Now let’s find the URL that starts the dashboard request.”

Step 3 — Start at the browser route (about 1 minute)

Open: `routes/web.php`.

Find/show: `Route::middleware(['auth', 'verified'])`, then the `GET /dashboard` route that points to `DashboardController::index`. Also point out the department/activity routes and `GET /report`, but stay on the dashboard route for the main trace.

Tell them: “A route connects a URL and HTTP action to PHP code. When an authenticated browser asks for `/dashboard`, Laravel sends the request to `DashboardController::index`. The routes are grouped behind authentication middleware; the email-verification caveat is explained later, so I will not claim verification is effectively enforced without checking the User model.”

Move on: “The route hands the request to the controller.”

Step 4 — Follow the dashboard controller (about 1½ minutes)

Open: `app/Http/Controllers/DashboardController.php`.

Find/show: `index()`, `Department::with('activities')->get()`, the loop that calls `calculateDepartmentStaffing($dept)`, the totals, and the final `return view('dashboard', [...])`.

Tell them: “The controller coordinates the request. It loads each department and its activities, asks the calculation service for a result, totals the facility numbers, and passes prepared data to the dashboard view. It is not where the WISN formula is written.”

Teaching check: Ask the learner, “Which line calls the maths?” They should point to `calculateDepartmentStaffing($dept)`.

Move on: “Before we inspect that maths, let’s see how the inputs get into the database.”

Step 5 — Show data entry and validation (about 1½ minutes)

Open: `app/Http/Controllers/DepartmentController.php`, then `app/Http/Controllers/WorkloadActivityController.php`.

Find/show: In `DepartmentController`, find `validationRules()`, `store()`, and the net-working-days check. In `WorkloadActivityController`, find `validationRules()` and `store()`; point out the allowed activity types and the rule that requires annual volume for `health_service` and `additional` activities.

Tell them: “These controllers accept form submissions and validate them before saving. Department inputs include the current headcount and the components used for AWT. An activity is created through its department relationship, so it is linked to the correct department. Invalid or incomplete entries should be rejected rather than silently used.”

Move on: “Those PHP model relationships describe how the saved records connect.”

Step 6 — Explain the models and database tables (about 2 minutes)

Open: `app/Models/Department.php`, then `app/Models/WorkloadActivity.php`.

Find/show: `Department::activities()` (`hasMany`), `WorkloadActivity::department()` (`belongsTo`), and `availableWorkingTimeHours()` in `Department.php`.

Tell them: “A department can have many workload activities, while each activity belongs to one department. AWT is calculated by the Department model from working days minus holidays, annual leave, sick leave, and training, multiplied by hours per day. In the current model, AWT is a computed attribute, not a manually entered total column.”

Then open: `database/migrations/2026_05_18_044256_create_departments_table.php`, `database/migrations/2026_05_18_044256_create_workload_activities_table.php`, and `database/migrations/2026_05_18_add_awt_breakdown_to_departments.php`.

Important teaching detail: the first department migration contains the earlier `available_working_time_hours` column. The later AWT-breakdown migration removes that column and adds the component fields. Explain that migrations run in order, so the final schema is understood by reading the later migration too—not by stopping at the first file.

Move on: “Now the service can read the department’s derived AWT and its activities.”

Step 7 — Teach the calculation service (about 3 minutes)

Open: `app/Services/WisnCalculatorService.php`.

Find/show: `calculateDepartmentStaffing(Department $department)`. Trace the code in this order:

1. `$department->available_working_time_hours` — gets the model's AWT value.
2. The loop over `$department->activities` — handles each activity type.
3. `health_service` — calculates standard workload, then required FTE from annual volume.
4. `support` — divides support hours by working hours per day and accumulates the support fraction.
5. `additional` — adds annual activity hours, then divides by AWT to get AAF FTE.
6. The CAF guard and formula for total required staff.
7. The WISN ratio, status method, and returned result array.

Tell them: "This service is the calculation engine. Health-service tasks, support time, and additional activities follow different code branches. The final required staff and ratio are assembled in one result array, rounded for display, and returned to the caller; this method does not save a WISN result row to the database."

Use the worked example: return to Section 8 of this guide and calculate the labelled example on paper. Keep saying "this is the formula currently coded here," not "this proves every methodological assumption has been clinically validated."

Move on: "The controller passes this result array to Blade, which turns it into the screen."

Step 8 — Show how the dashboard displays the result (about 1½ minutes)

Open: `resources/views/dashboard.blade.php`.

Find/show: "Facility Staffing Dashboard," the summary cards, the department table, the two `<canvas>` chart elements, and the script near the bottom that turns `@json($departmentResults)` into Chart.js datasets. Point to the report link if visible.

Tell them: "Blade is the server-side HTML template. The controller has already prepared the numbers; this view displays them as cards and a department table. The JavaScript reads the prepared results and Chart.js draws the charts. The view displays the calculation—it is not a second calculation engine."

Move on: "The downloadable report uses the same calculation service, rather than a separate WISN formula."

Step 9 — Follow the PDF report path (about 1 minute)

Open: `app/Http/Controllers/ReportController.php`, then `resources/views/reports/wisn-summary.blade.php`.

Find/show: `generate()`, the call to `calculateDepartmentStaffing()`, `Pdf::loadView('reports.wisn-summary', ...)`, and the download response. In the Blade report, point to the facility totals, department-level results, activity breakdown, and formula summary.

Tell them: "The report controller calls the same WISN service used by the dashboard, adds the AWT input breakdown, and passes the data to a Blade report template. DomPDF renders that template and Laravel returns a PDF download."

Transition: "That finishes the annual planning path. The next files are a different feature with manually entered shift-level data."

Part B — trace the separate inpatient census and staffing view

Step 10 — Show the operations route and daily timeline (about 1 minute)

Open: `routes/web.php` and find `GET /operations` and `POST /operations/snapshots`. Then open `resources/views/operations/index.blade.php`.

Find/show: the date picker, the three handover cards for 12:00 AM / 7:00 AM / 7:00 PM, the inpatient-unit table, and the daily timeline below it. Point out that there is no OPD row and that Emergency is labelled for admitted/observation-bed counts.

Tell them: "This is separate from annual WISN. The midnight row is a point-in-time census only. At the 7 AM and 7 PM handovers, the operator records both patients actually observed and nurses who actually came on duty. Each census is entered directly; the application does not infer transfers or admissions that were not recorded."

Move on: “The form posts those fixed date-and-shift counts to a controller for validation.”

Step 11 — Explain snapshot validation (about 1 minute)

Open: `app/Http/Controllers/OperationalSnapshotController.php`.

Find/show: `store()`, the date and shift rules, complete inpatient-unit ID check, required patient count, and the conditional on-duty rule for 7 AM / 7 PM. Point out that there is no manually entered required-nurse field.

Tell them: “The controller accepts a patient count for every inpatient nursing unit. It requires an actual on-duty nurse count for the day and night handovers, but does not ask for one at midnight. OPD is filtered out. It then passes only validated values into the service; this is manual entry, not an EHR or attendance import.”

Move on: “The required count, when available, comes from a separate, configured standards lookup—not from a fixed number hidden in the form.”

Step 12 — Show standards configuration and calculation (about 2 minutes)

Open: `app/Http/Controllers/OperationalStaffingStandardController.php`, `app/Models/OperationalStaffingStandard.php`, and `resources/views/operations/standards.blade.php`. Then show the admin-only standards routes in `routes/web.php` and `app/Http/Middleware/EnsureUserIsAdmin.php`.

Find/show: unit type, day/night shift, patients per nurse, optional minimum, source/version, effective dates, active state, approver, and approval timestamp. Then show `currentFor()` and the formula in `recordRound()`.

Tell them: “No default ratio is seeded. An administrator must enter and activate a locally approved source-backed target. For an effective standard, required nurses equal the larger of the approved minimum and the ceiling of observed patients divided by patients per nurse. If no standard applies, the system stores the census and on-duty count but leaves the required count blank.”

Move on: “The service can compare units only when they have calculated targets, and the result is still only a human-reviewed possibility.”

Step 13 — Trace suggestions, storage, and tests (about 2 minutes)

Open: `app/Services/OperationalStaffingService.php`.

Find/show: `recordRound()`, `DB::transaction()`, `currentFor()`, `createRecommendations()`, then `approve()`, `ensureLatestRound()`, and `ensureFresh()`.

Tell them: “The service saves an observed round and unit snapshots in a transaction. It compares actual on-duty counts with the applicable calculated targets and can propose an aggregate count between units. Approval records a review only—it does not select a nurse, change the snapshot, alter department headcounts, update a roster, or affect WISN.”

Then open: `database/migrations/2026_10_06_000001_add_inpatient_shift_census_and_staffing_standards.php` and the existing operational migrations. Show the standard table, date/shift/time fields, calculated requirement and standard link, and separate mobilization recommendation table.

Finally open: `app/Http/Controllers/OperationalDashboardController.php`, return to `resources/views/operations/index.blade.php`, and show `tests/Feature/OperationalStaffingTest.php`. Point out the previous-day view and that tests cover target math, midnight-only census, OPD exclusion, missing standards, human approval, and WISN separation.

Tell them: “Tests show intended behavior; they are not proof that tests ran successfully on this machine. Run `php artisan test` in a configured Laravel environment before claiming a passing test run.”

Finish the walk-through (about 1 minute)

Close the editor and return to the dashboard or your presentation slide. Say:

“The annual WISN path is entered department/activity data → controller → calculation service → dashboard or PDF. The operational path is a separate manual shift snapshot → aggregate suggestion → human review. The next engineering steps are validating local standards, improving access control, and connecting trustworthy data sources safely; the current prototype does not make clinical staffing decisions automatically.”

Ask the learner to trace both paths back without reading. If they can say what each file does and what the system does **not** do, they are ready to present.

README warning: the root `README.md` is still the stock Laravel starter README. Use `ARCHITECTURE.md`, the project report, and the code—not the starter README—as the project-specific sources.

13. Local run and the current empty-database setup

A typical development machine needs PHP 8.1+, Composer, Node/npm, and a database configured in `.env`. This user's current test database is already migrated and has a single administrator account; there is no need to migrate or seed again before following Section 14.

If you need to start the local web process, run:

```
php artisan serve
```

In another terminal, if browser assets are not already built, run:

```
npm install
npm run dev
```

Open the URL printed by `php artisan serve`, then visit `/login`.

For the current manual-from-zero test: do not run `php artisan migrate:fresh` again and do not run `php artisan db:seed.migrate:fresh`. `migrate:fresh` deletes all database rows before recreating tables.

`DatabaseSeeder` calls the demo admin seeder, then deletes workload activities and departments before inserting its sample departments and activities. Running it would replace the clean manual-data exercise with demo data. No staffing ratios are seeded in any case.

If you intentionally want a disposable sample-data database later, use `php artisan db:seed` only after confirming the database contains no data you need. The `AdminUserSeeder` alone creates/updates the demo account `admin@wisn.org` and resets its password to the repository's demo password `password`; it does not add departments or activities. Do not use that password outside local demonstration, and do not re-run that seeder if you have already changed the admin password and want to keep it.

The application menus are rendered by `resources/views/layouts/navigation.blade.php`. The primary order for a manual clean-data test is: **login → Departments → Activities → Dashboard → Report/PDF → Operations → Profile → logout**. Operations target configuration is an admin-only subpage. The separate click-by-click and code-path test is in the next section.

Seed warning: the current `DatabaseSeeder` is destructive to department/activity data. Never run it against real hospital or operational data. It also does not create staffing standards; local approval and entry are required before ratios are used.

Demo order (about 4-6 minutes)

1. **Start with the dashboard.** Explain current staff, required staff, ratio, status, and chart.
2. **Open Departments.** Show one department's AWT components; explain AWT is derived, not manually stored as one number.
3. **Open its Activities page.** Show the three categories and explain annual volume/time standard.
4. **Return to Dashboard.** Point out that changing saved inputs causes the calculation to be recomputed on the next request.
5. **Open Report.** Download the PDF and show that it summarizes the same calculation service.
6. **Open Operations.** Show the date picker and the 12:00 AM / 7:00 AM / 7:00 PM timeline. Enter only clearly labelled demo counts; never enter patient identifiers. If the facility has a validated local target, have an administrator configure its source, version, effective dates, and approval before demonstrating the calculation. If no target has been validated, show the blank requirement and explain that the application correctly does not invent one. Only show an aggregate suggestion if configured targets support it; explain that it is for human review, not a nurse assignment.

7. **Review a prior date.** Use the previous-day control to show how each handover is stored separately within the midnight-to-midnight day.
8. **Close with limitations and next steps.** This demonstrates technical confidence rather than hiding prototype boundaries.

Emergency counts must be admitted/observation-bed patients only; OPD and walk-in Emergency visits are excluded. If a unit is missing from a complete submission, reload and save all inpatient nursing units.

14. Hands-on walkthrough: test from an empty database

This is the practical lab to follow in the browser. It assumes the database has already been migrated and contains only the administrator account, as in the current test setup. It deliberately creates data through the user interface so you can see how each form maps to Laravel code.

Safe test-data rule: use the sample workloads and operational standards below only in a disposable local practice database. They are teaching numbers, not clinically approved workload volumes or patient-to-nurse ratios. In a real facility, enter only locally verified data and an approved staffing standard. Do not enter patient names or other identifiers.

Step 0 — Confirm the starting point

Before clicking around, check the premise:

- You can log in with the administrator account that already exists.
- There are no departments or activities yet, and no operational standards are configured.
- You already ran `php artisan migrate:fresh`; do not run it again during the walkthrough.
- Do not run `php artisan db:seed` for this manual exercise. `DatabaseSeeder` deletes department and activity rows and then inserts a sample data set. `AdminUserSeeder` alone is also unnecessary if the admin already exists; rerunning it resets that demo account's password.

If the web app is not running, start it from the repository root with `php artisan serve`. Open the URL printed in the terminal. If Vite assets are not built, run `npm run dev` in another terminal. These commands do not reset the database.

Step 1 — Log in and arrive at the dashboard

1. Open `/login` and enter the existing admin credentials.
2. The browser first makes `GET /login`. `routes/auth.php` maps it to `AuthenticatedSessionController@create`, which returns `resources/views/auth/login.blade.php`.
3. Submitting the form sends `POST /login`. `AuthenticatedSessionController@store` calls `LoginRequest::authenticate()`, regenerates the session ID, and redirects to the intended page. The normal home path is `/dashboard` (`RouteServiceProvider::HOME`).
4. `GET /dashboard` is registered in `routes/web.php` as `dashboard`. It passes through the authenticated application middleware and reaches `DashboardController@index`.
5. `DashboardController` loads departments and their activities. With none present, the page should show an empty/zero-data state. It cannot calculate a meaningful department requirement until data is added.

Show in the editor: `routes/auth.php` → `app/Http/Controllers/Auth/AuthenticatedSessionController.php` → `app/Http/Requests/Auth/LoginRequest.php` → `app/Providers/RouteServiceProvider.php` → `routes/web.php` → `app/Http/Controllers/DashboardController.php`.

Say: “The login page is a GET request. The form submits a POST. Laravel checks the credentials, refreshes the session, and sends me to the dashboard. The dashboard is empty because we have not created any departments yet.”

Step 2 — Open Departments and inspect the empty list

1. Use the navigation link **Departments**. It points to `route('departments.index')`, which is `GET /departments`.
2. `Route::resource('departments', DepartmentController::class)` in `routes/web.php` maps the request to `DepartmentController@index`.
3. The controller runs `Department::withCount('activities')->get()` and passes the result to `resources/views/departments/index.blade.php`.
4. With no rows, the view offers **Add First Department**. No department or activity row is created merely by opening the page.

Show: the navigation link, the resource route, `DepartmentController@index`, the `Department` model, and the empty-state branch in `departments/index.blade.php`.

Step 3 — Add the first department

Click **Add First Department** or **Add Department**. This is `GET /departments/create`, named `departments.create`. `DepartmentController@create` sends the available Operations classifications to `resources/views/departments/create.blade.php`.

The form has two separate “type” concepts. Their source and purpose are different:

Form field	Where the options/data originate	What it means
Department Name	Text typed by the user; saved in <code>departments.name</code> .	A readable name, such as “Medical Ward.”
Department Type (<code>type</code>)	The select options are an array written directly in the Blade form: <code>Inpatient - Standard, Inpatient - High Acuity, Outpatient, Emergency, Surgical/OT</code> .	A broad WISN/report grouping. It is not the key used to look up an Operations target. The current controller validates this as a required string; it does not validate against that Blade list.
Operations Unit Classification (<code>operational_unit_type</code>)	<code>DepartmentController</code> passes <code>Department::OPERATIONAL_UNIT_TYPES</code> from <code>app/Models/Department.php</code> to the view.	A code used by Operations to distinguish General Ward, Surgical Ward, ICU, PICU, NICU, Emergency admitted/observation beds, OPD, and other units. Outpatient is excluded from Operations.
Current Staff Headcount (WISN)	A whole-number input saved as <code>departments.current_staff</code> .	Annual/establishment headcount for WISN. It is not the number of nurses actually present at a 7 AM or 7 PM handover.
Working Days/Year	An editable form input; create form starts at 260.	Gross available workdays in the selected annual reference period.
Public Holidays	Editable input; default 13.	Days deducted from gross workdays.
Annual Leave Days	Editable input; default 18.	Days deducted from gross workdays.
Sick Leave Days	Editable input; default 12.	Days deducted from gross workdays.
Training Days	Editable input; default 5.	Days deducted from gross workdays.
Working Hours/Day	Editable input; default 8.	Hours per working day.

Form field	Where the options/data originate	What it means
Available Working Time (AWT)	Not typed into a separate stored AWT field. The six values above are stored; <code>Department::availableWorkingTimeHours</code> accessor computes AWT.	Annual hours available for one nurse. The form's JavaScript preview is only a convenience; the server-side PHP accessor is used by calculations.

The defaults are examples that produce:

$(260 - 13 - 18 - 12 - 5) \times 8 = 1,696$ hours per nurse per year

They can be changed to match an approved planning method. The request is validated by `DepartmentController::validationRules()`. The controller also checks that leave/holiday/training deductions leave positive net working days. If the `type` is exactly `Outpatient`, it forces `operational_unit_type=outpatient`; if exactly `Emergency`, it forces `emergency_inpatient`.

For a training-only example, make these three departments so we can test both WISN and the Operations filter:

Department Name	Department Type	Operations Unit Classification	WISN headcount	AWT inputs
Medical Ward	Inpatient - Standard	General inpatient ward (<code>general_ward</code>)	2	Leave the example defaults for this first example.
Surgical Ward	Surgical/OT	Surgical ward (<code>surgical_ward</code>)	1	Same example defaults.
Outpatient Department (OPD)	Outpatient	Outpatient Department (<code>outpatient</code>)	4	Same example defaults.

These headcounts are invented only to exercise the screens. For an inpatient nursing unit, pick the appropriate unit classification even if the broad Department Type is different. The OPD row is useful for confirming that it appears in WISN department management but not in Operations.

When you click **Save Department**, the browser sends `POST /departments` (named `departments.store`) with a CSRF token. `DepartmentController@store` validates the request, performs its net-AWT-days check and the Outpatient/Emergency mapping, then calls `Department::create($validated)`. Eloquent inserts one row in `departments`. The controller redirects to the department list with a success message. The page itself does not store the computed AWT number; it stores the six AWT inputs.

Exact code trace: `routes/web.php` → `DepartmentController@create` / `DepartmentController@store` → `Department::OPERATIONAL_UNIT_TYPES` and `Department::create()` → `departments` table → `resources/views/departments/create.blade.php` / `departments/index.blade.php`.

Say: “The Type list is a broad WISN label written in the Blade form. The Operations classification comes from a PHP constant in the Department model. Current Staff Headcount feeds annual WISN. The 1,696-hour AWT is calculated from the six stored fields, not copied from the browser preview.”

If validation fails, Laravel returns to the form with errors and old input. The server rechecks everything even if the browser already displayed an AWT preview.

Step 4 — Open a department’s Activities page

Return to `GET /departments`. Each row has an **Activities** action. It links to `route('activities.index', $department->id)`, a `GET /departments/{department}/activities` route registered explicitly in `routes/web.php`.

`WorkloadActivityController@index` receives the `Department` through implicit route model binding, loads `$department->activities`, and returns `resources/views/activities/index.blade.php`. The page shows the

relationship's activity list (empty at first) and a WISN setup stepper. For a new department, the stepper highlights adding a health-service activity first.

Show: `routes/web.php`, `WorkloadActivityController@index`, `Department::activities()`, `WorkloadActivity`, and the page's stepper in `resources/views/activities/index.blade.php`.

Step 5 — Add activities (use these fictional practice values)

Add one activity of each type to **Medical Ward**. The figures below exist only to make a visible dashboard calculation; they are not clinical standards.

Activity name	Activity type	Time standard	Annual volume	Why this example is entered this way
Practice direct-care task	Health Service (<code>health_service</code>)	0.50 hours	6,000	Direct service workload uses time per occurrence and annual volume.
Practice shift handover	Support (<code>support</code>)	0.50 hours per shift	Leave blank	The code uses support time as a fraction of each working shift; no annual-volume count is required.
Practice teaching task	Additional (<code>additional</code>)	0.25 hours	160	Additional allowance uses annual hours (<code>volume × time</code>) in the current service.

To save an activity, fill in the form and click **Add Activity**. The browser sends `POST /departments/{department}/activities` (for example, named `activities.store`).

`WorkloadActivityController@store` validates the name, type, time, and required annual-volume rule. It creates the record through `$department->activities()->create($validated)`, which sets the department foreign key. Eloquent inserts into `workload_activities`; the controller redirects to that department's activities page.

The category options are rendered by the activity Blade page and constrained by the database enum/controller rule to `health_service`, `support`, and `additional`. Time standards must be numeric and at least 0.01 hours. Annual volume must be an integer of at least 1 for Health Service and Additional; it may be blank for Support. A support time standard is treated by the service as hours per shift, even though the form's generic label says hours per occurrence—explain this special case.

Edit path: click Edit → `GET /departments/{department}/activities/{activity}/edit` → `WorkloadActivityController@edit` → `resources/views/activities/edit.blade.php`. Save the edit → the form sends `PUT /departments/{department}/activities/{activity}` using `@method('PUT')` → `WorkloadActivityController@update` validates and calls `$activity->update($validated)`.

Delete path: submit the row's delete form → `DELETE /departments/{department}/activities/{activity}` using `@method('DELETE')` → `WorkloadActivityController@destroy` → `$activity->delete()`. The page reloads without that activity. Deleting an activity changes future calculations but does not create a separately stored WISN result.

Training checks: try submitting a Health Service activity without an annual volume; the controller should reject it. Then add a Support activity with the volume blank; that is allowed. The activity list and stepper should update after a successful save.

Step 6 — Return to the annual WISN Dashboard

Click **Dashboard**. The full route-to-result path is:

```
GET /dashboard (route name dashboard)
→ DashboardController@index
```

```

→ Department::with('activities')->get()
→ WisnCalculatorService::calculateDepartmentStaffing() for each department
→ aggregate totals and facility ratio
→ resources/views/dashboard.blade.php

```

The service reads AWT from the Department accessor and the saved `WorkloadActivity` rows. It calculates HS standard workload and FTE, support allowance (CAF), additional FTE, total required staff, and the WISN ratio. The calculation is performed when the page is requested; no WISN-results table is written. Department edits and activity additions therefore affect the next dashboard/report calculation.

For **Medical Ward** with the practice values above:

```

AWT = 1,696 hours
HS standard workload = 1,696 / 0.50 = 3,392 occurrences/nurse/year
HS FTE = 6,000 / 3,392 ≈ 1.7689
Support fraction = 0.50 / 8 = 0.0625; CAF = 1 / (1 - 0.0625) ≈ 1.0667
Additional FTE = (160 × 0.25) / 1,696 ≈ 0.0236
Required = (1.7689 × 1.0667) + 0.0236 ≈ 1.91 nurses
WISN ratio = 2 current / 1.91 required ≈ 1.05

```

The Medical Ward row should show about **1.91 required** and **1.05 WISN ratio**, with status `surplus` because the unrounded ratio is greater than 1. Other departments with no activities have no meaningful requirement (`no_data` in the service). The facility total is calculated from total current divided by total required—not by averaging departmental ratios.

Show: `DashboardController@index`, `WisnCalculatorService::calculateDepartmentStaffing()`, the `Department` AWT accessor, `WorkloadActivity` relation, and `resources/views/dashboard.blade.php`.

Step 7 — Generate and inspect the PDF report

On the Dashboard click **Download Report** (or the report link in navigation). This is `GET /report`, named `report.generate`.

1. `ReportController@generate` loads departments and activities.
2. It calls the same `WisnCalculatorService` used by the Dashboard.
3. It adds the six input values as an AWT breakdown and calculates facility totals.
4. `Pdf::loadView('reports.wisn-summary', $data)` renders `resources/views/reports/wisn-summary.blade.php` through DomPDF.
5. Laravel returns a download named `WISN_Facility_Staffing_Report.pdf`.

The report is generated from current data at download time; it does not write another calculation row. It is an **annual WISN report**, not an Operations census report. Open the downloaded file and compare Medical Ward's requirement and ratio to the Dashboard.

Show: the Dashboard link → `routes/web.php` → `ReportController@generate` → `WisnCalculatorService` → `resources/views/reports/wisn-summary.blade.php` → PDF download.

Step 8 — Open Operations and understand the empty standard state

Click **Operations** in the main navigation. This sends `GET /operations` to `OperationalDashboardController@index`; it selects the facility-local date and shift, filters out `outpatient`, loads the latest snapshots, applicable standards, and recommendations, then renders `resources/views/operations/index.blade.php`.

- Medical Ward and Surgical Ward should appear because they are inpatient units.
- OPD should not appear in the Operations entry table.
- The initial target-standard list is empty; the app should say that no requirement will be calculated until an approved standard is configured.
- A 12:00 AM selection is census-only. 7:00 AM and 7:00 PM also ask for nurses who actually came on duty.

The `operational_unit_type`, not the broad Department Type field, determines whether a department is OPD and which standard lookup applies. Emergency is intended for admitted/observation-bed patients, not walk-ins.

Show: `OperationalDashboardController@index`, `OperationalRound::SHIFTS`, the query that excludes `outpatient`, and the empty/target-warning states in `resources/views/operations/index.blade.php`.

Step 9 — Configure test-only standards as the administrator (optional)

Skip this step if you only want to test census capture. With no standard, Operations still saves patient and on-duty counts but correctly leaves required nurses blank and produces no target-based proposal.

To demonstrate calculations and a count recommendation, only use the following invented standards in an isolated practice database. **Do not use these ratios as hospital policy.**

1. On the Operations page select **Configure target ratios**. The link is shown to `is_admin` users.
2. `GET /operations/standards` passes through `auth`, `verified`, and `admin` middleware. `EnsureUserIsAdmin` checks `users.is_admin`; `OperationalStaffingStandardController@index` supplies inpatient unit labels, supported day/night shifts, and saved standards to `resources/views/operations/standards.blade.php`.
3. Enter a day standard for General inpatient ward: `patients per nurse = 4`, minimum `1`, source name `TRAINING ONLY – not a clinical policy`, version `demo`, effective-from date equal to the selected census date, no end date, activate/approve checked.
4. Add another day standard for Surgical ward: `patients per nurse = 3`, minimum `1`, with the same clear training-only source label and date.
5. Save each. `POST /operations/standards` validates the fields and writes through `OperationalStaffingStandardController@store` and the `OperationalStaffingStandard` model. If activated, the authenticated admin ID and time are stored as approval metadata. This is a demonstration of the software's approval mechanism, not evidence of clinical approval.
6. Return to Operations and confirm that the unit/shift targets and source/version are visible. Midnight never uses these standards.

The real application intentionally does not ship with a default patient-to-nurse ratio. In a real facility, only the locally governed, source-backed ratio should be activated.

Step 10 — Enter midnight census

Select the current facility date and **Midnight census (12:00 AM)**. Enter a directly observed patient count for every inpatient unit in the table and save.

- This submits `POST /operations/snapshots (operations.snapshots.store)`.
- `OperationalSnapshotController@store` validates date, shift, all unit IDs, patient counts, and notes. On midnight it does not require or save on-duty staff.
- `OperationalStaffingService::recordRound()` stores the `OperationalRound` and one `DepartmentOperationalSnapshot` per inpatient unit inside a transaction. It sets the scheduled observation time to midnight in `OPERATIONS_TIMEZONE`.
- The `OperationalRound` and snapshot models connect the date/shift to units, user, and historical target reference.
- The controller redirects to the selected date/shift on Operations.

Midnight is an exact observed census, not a count derived from admissions or discharges. No required staffing or move suggestion should be produced for the midnight round.

Step 11 — Enter 7 AM census and actual on-duty counts

Choose **Day handover (7:00 AM)**. Enter the patient census and actual nurses who came on duty for **every inpatient unit**, then save. OPD is not part of the form. The controller checks the unit list submitted by the browser exactly matches the current inpatient-unit set; if it changed while the form was open, reload the page and enter the complete set.

Use these figures only with the two training-only standards from Step 9:

Inpatient unit	Patients observed	Nurses actually on duty	Training target	Calculated requirement	Balance
Medical Ward	4	3	1 nurse per 4 patients; min 1	1	2 above target
Surgical Ward	9	1	1 nurse per 3 patients; min 1	3	2 below target

The formula is `max(minimum nurses, ceil(patients / patients per nurse))`. When the standards apply, the service can create a pending aggregate suggestion for 2 staff from Medical Ward to Surgical Ward. This is only a software demonstration; it is not a safe real-world instruction.

Code trace: `POST /operations/snapshots` → `OperationalSnapshotController@store` →

`OperationalStaffingService::recordRound()` → `OperationalStaffingStandard::currentFor()` → snapshots and recommendations → redirect back to Operations.

Step 12 — Understand 7 PM and recommendation review

The **Night handover (7:00 PM)** uses the same route and validation but is a separate date/shift round. Enter a new direct patient census and actual night-shift on-duty count for every inpatient unit. It does not carry day counts forward automatically.

If a current recommendation appears:

1. Read its source unit, destination unit, count, rationale, status, and freshness message.
2. A user with an authenticated session may submit approve or decline with an optional note. The relevant POST route reaches `MobilizationRecommendationController`, which calls `OperationalStaffingService`.
3. Approval is recorded only after the service confirms the recommendation is still pending, the round is the latest complete staffed handover, it is within the configured freshness window, the inpatient-unit set is unchanged, the saved staffing standards still apply, and the proposed counts do not breach source or destination requirements.
4. The review writes reviewer/time/note and status to the recommendation. It does not alter on-duty headcount, department `current_staff`, schedule, or roster.
5. A new round or a staffing-standard status/version change supersedes pending recommendations.

Important timing: freshness is measured from the scheduled `observed_at` time (7:00 AM or 7:00 PM), not from when you typed the form. With the default 30-minute setting, a 7:00 AM round is considered stale at 7:30 AM. A late-entered historical round may still be saved, but its move recommendation cannot be approved as fresh. For a timely approval test, arrange the demo close to a handover. Any larger freshness setting should be confined to a local test environment and restored before real use.

Step 13 — Review history and distinguish the three staffing concepts

Use the date picker/timeline to review the latest midnight, day, and night records for that date. If a census is corrected and saved again, the latest revision is displayed while older rounds remain stored. A new department/unit after the round changes the required complete-unit set and blocks recommendation approval until a full new round is recorded.

Keep these values separate when explaining the screen:

- `departments.current_staff`: annual WISN planning headcount.
- `department_operational_snapshots.on_duty_staff`: actual nurses present at the selected 7 AM or 7 PM handover.
- `calculated_required_staff`: target-based shift requirement for a specific census/round, only when an applicable approved target exists.

None is a substitute for another. Operations does not update the WISN dashboard totals.

Step 14 — Open Admin Profile, User Profile, and log out

There is no separate “Admin Profile” page. An administrator uses the same **Profile Settings** page as any other signed-in user; the `is_admin` flag only reveals the target-configuration link and authorizes the `/operations/standards` routes.

1. Open the user menu and click **Profile**. GET `/profile` → `ProfileController@edit` → `resources/views/profile/edit.blade.php`, which includes three partial forms.
2. **Profile information:** save name/email. The form sends PATCH `/profile`; `ProfileController@update` uses `ProfileUpdateRequest`, fills the current user, clears `email_verified_at` if email changes, saves, and redirects back. It does not edit `is_admin`.
3. **Password:** enter current/new/confirmation. The form sends PUT `/password` to Breeze's `Auth\PasswordController@update` and shows validation/status feedback.
4. **Delete account:** the form sends DELETE `/profile` to `ProfileController@destroy`; it requires the current password, logs out, deletes the user, and invalidates the session. **Do not test account deletion on the only administrator**, or you may remove your only admin login.
5. Optional: `/register` uses Breeze's `RegisteredUserController`; newly registered users receive the database default `is_admin=false`. There is no application screen to promote them to administrator. The current User model also does not implement `MustVerifyEmail`, so treat the `verified` middleware behavior as a caveat, not as guaranteed email enforcement.
6. Use the navigation logout action. POST `/logout` calls `AuthenticatedSessionController@destroy`, logs out, invalidates the session, and returns to `/`.

Show: `routes/web.php`, `routes/auth.php`, `ProfileController`, `ProfileUpdateRequest`, the three `resources/views/profile/partials/` forms, `User.php`, and `EnsureUserIsAdmin.php`. Explain that registration/profile fields do not include an admin-role switch.

Final verification checklist for this clean-data run

- [] I did not run `migrate:fresh` or `DatabaseSeeder` again after beginning manual entry.
- [] I created at least two inpatient units and one optional OPD test department.
- [] I know where the Department Type choices and Operations classification choices are defined.
- [] I can explain how the six AWT values produce 1,696 hours in the example.
- [] I added one Health Service activity, one Support activity, and one Additional activity and can state why Support has no annual volume.
- [] The dashboard updated after the saved data; I can trace its route, controller, service, model relations, and view.
- [] I downloaded the WISN PDF and can tell that it is generated from current WISN data, not Operations.
- [] I saw that Operations excludes OPD and uses direct census at midnight and the two handovers.
- [] I left targets blank unless I deliberately used clearly labelled training-only standards in a disposable database.
- [] I understand that an approved move is still only an aggregate human review and changes no nurse roster or actual counts.
- [] I checked the same Profile Settings page as admin and understand why it is not a separate Admin Profile page.
- [] I did not delete the only administrator account.

Presenter file order for this practical walkthrough

Order	Open this file	Point out / say
1	<code>routes/auth.php</code> , <code>AuthenticatedSessionController.php</code> , <code>LoginRequest.php</code>	GET login displays the form; POST authenticates and redirects.
2	<code>routes/web.php</code> , <code>DepartmentController.php</code> , <code>Department.php</code>	Department list/create/store; explain both type fields, AWT input storage, accessor, and validations.

Order	Open this file	Point out / say
3	<code>resources/views/departments/create.blade.php</code>	<code>Show.php</code> , coded Department Type options, Model-provided Operations options, input defaults, live preview.
4	<code>WorkloadActivityController.php</code> , <code>WorkloadActivity.php</code> , <code>WisdCalculatorService.php</code>	Nested activity routes, validation, relation create/update/delete, calculation inputs.
5	<code>DashboardController.php</code> , <code>dashboard.blade.php</code>	Per-department service call, facility aggregation, chart/table output.
6	<code>ReportController.php</code> , <code>reports/wisd-summary.blade.php</code>	Same service, DomPDF template, download response.
7	<code>OperationalDashboardController.php</code> , <code>OperationalSnapshotController.php</code> , <code>OperationalStaffingService.php</code>	Three census times, full-unit validation, target lookup, saved snapshot, aggregate proposal.
8	<code>OperationalStaffingStandardController.php</code> , <code>OperationalStaffingStandard.php</code> , <code>EnsureUserIsAdmin.php</code>	<code>Admin.php</code> , only target configuration and source/effective/approval metadata.
9	<code>MobilizationRecommendationController.php</code> , <code>MobilizationRecommendation.php</code>	Human review and audit; no staffing or roster mutation.
10	<code>ProfileController.php</code> , <code>ProfileUpdateRequest.php</code> , auth profile partials, <code>User.php</code>	One profile for admins and users; profile/password/delete behavior and role boundary.

Teaching sentence to repeat: “The page form collects input; the route chooses a controller; the controller validates and coordinates; a model relationship writes or reads database rows; a service calculates business results; and Blade displays the response. For the PDF route, DomPDF renders a Blade template and returns a download.”

15. Security, testing, and quality questions

What the code does

- Routes are grouped under Laravel's `auth` and `verified` middleware names.
- Controllers validate input before saving it.
- Laravel web forms use CSRF protection through the web middleware stack.
- Eloquent parameterizes database operations; Blade's `{{ }}` output is escaped by default.
- WISN logic is separated into a service and now has a regression test.
- The operational approval is transaction-based and checks that a count move does not cross recorded targets.
- Operational text inputs warn users not to store patient identifiers.

Important access-control caveat

The `is_admin` flag and `admin` middleware restrict target-standard configuration to an administrator. Operational census entry and aggregate recommendation review are available to authenticated users; this is not a granular nursing-manager role system. The `AdminUserSeeder` creates a demo administrator, so its demo password must be changed and the account secured before deployment.

The current `User` model has its `MustVerifyEmail` implementation commented out. Laravel's `verified` middleware only enforces verification for models that implement the verification contract. Therefore, do not claim email verification is effectively enforced without checking/fixing that model behavior.

Tests

The repository includes standard Breeze authentication/profile tests, a WISN regression test, and operational feature tests for ratio-based required counts, midnight census-only behavior, OPD exclusion, missing standards, administrator-only configuration, recommendations, human approval, standard-change invalidation, stale rounds, superseding, and department-history protection. Run:

```
php artisan test
```

A test file existing is not the same as a passing test run. Before the presentation, run the suite on a machine with dependencies and a working database driver; report honestly whether it passed.

16. Evaluator questions and ready-to-say answers

Use the answer in quotes as the first response. Add detail only if the evaluator asks for it.

Project and method

Q1. What is WISN Core?

"It is a Laravel web prototype that calculates and displays nursing staffing estimates by hospital department using entered workload volumes, time standards, and available working time."

Q2. What problem does it solve?

"It makes selected WISN calculations repeatable and easier to interpret than doing each calculation manually in a spreadsheet. It gives managers a dashboard and report for reviewing staffing estimates."

Q3. What does WISN stand for?

"Workload Indicators of Staffing Need, a WHO methodology for estimating health-worker requirements from workload and time standards."

Q4. Is the application itself the WHO methodology?

"No. WISN is the methodology; this application is a software implementation of selected calculations and workflows. The underlying standards and assumptions still need appropriate expert validation."

Q5. Why is annual data used?

"The core WISN calculation compares annual workload with annual available working time. It is useful for workforce planning over a reference period, not for choosing a nurse to move this minute."

Q6. What does a WISN ratio of 0.8 mean?

"It means current staff is about 80 percent of the calculated requirement under the entered inputs. The app labels values below 0.90 as critical, but the ratio is an estimate—not an automatic clinical order."

Q7. What does a ratio above one mean?

“The department's entered current headcount is above its estimated workload-based requirement. It does not automatically mean that a particular nurse is safe or available to redeploy.”

Q8. Is the data real?

“The repository contains demo/seed data and reference material. The project scope describes simulated data. I would label demo values as simulated and cite a source before presenting any individual figure as official.”

Technology and architecture

Q9. What is Laravel?

“Laravel is a PHP web framework. It gives the project a structured way to define routes, controllers, database models, validation, authentication, and page templates.”

Q10. Why Laravel instead of plain PHP?

“Laravel provides standard tools for routing, database migrations, ORM access, validation, authentication, and templating, which makes a small application easier to organize and maintain.”

Q11. What is MVC?

“MVC separates concerns: models represent stored data, views render the interface, and controllers coordinate requests. The WISN math is additionally kept in a service class.”

Q12. What happens when someone opens the dashboard?

“The web route sends the request to `DashboardController`; it loads departments and activities, calls `WisnCalculatorService`, then passes the results to a Blade view that renders the cards, table, and charts.”

Q13. What is Eloquent?

“Eloquent is Laravel's ORM. It maps database rows to PHP model objects and defines relationships such as one department having many workload activities.”

Q14. What are migrations?

“Migrations are version-controlled database schema changes. They let a developer create or update tables in a repeatable way with `php artisan migrate`.”

Q15. Why use Blade?

“Blade renders HTML on the server and can display the data prepared by Laravel. That fits this CRUD-and-dashboard prototype without needing a separate JavaScript frontend application.”

Q16. What does Vite do?

"Vite serves front-end assets during development and builds CSS/JavaScript for deployment. The Laravel layout loads the app CSS and JavaScript through the Vite integration."

Q17. Which tool draws the charts?

"Chart.js draws the dashboard charts. In this project it is loaded from a CDN on the dashboard page."

Q18. How is the PDF report made?

"`ReportController` calls the same WISN calculation service, passes the results to a Blade report template, and DomPDF renders that template as a downloadable PDF."

Q19. Is it a single-page app or API-first system?

"No. The main interface is a server-rendered Laravel web application using Blade. The core user flow is in `routes/web.php`."

Data and calculation

Q20. Where is AWT stored?

"Its components are stored on the department record. The total is calculated by an Eloquent accessor, so it is not a separately entered AWT column in the current schema."

Q21. What is a time standard?

"It is the entered time associated with an activity. For health-service and additional activities it is time per occurrence; for support activities the code treats it as hours per shift."

Q22. How does the activity table relate to departments?

"A workload activity stores a department foreign key. One department can have multiple activities, and each activity belongs to one department."

Q23. Does the database store every calculated ratio?

"The dashboard calculation is generated when the request is processed and passed to the view. The calculated result is not stored as a separate WISN-results table in this implementation."

Q24. How is the facility ratio calculated?

"The dashboard divides the sum of current staff by the sum of required staff across departments. It is not a simple average of the department ratios."

Q25. Where does the demo data come from?

"The seeders contain the demo rows and comments about their assumptions and cited context. The bundled proposal, report, WHO material, and Nepal source documents provide references. I would verify a specific value against its cited source before calling it official."

Operations, safety, and limitations

Q26. Is this a live patient or staff feed?

"No. Counts are entered manually at 12:00 AM, 7:00 AM, and 7:00 PM. Operations is not connected to an EHR, patient census, or attendance roster."

Q27. How do you calculate the 7 AM or 7 PM required nurses?

"Only if an administrator has activated an applicable, effective, locally approved target. The calculation is the ceiling of observed patients divided by patients per nurse, with the configured minimum applied. Without a target, required nurses stay blank."

Q28. What ratio does the system use by default?

"There is no default ratio. The facility must configure a source-backed target for each inpatient unit type and day or night shift. Nepal standards may offer candidate references, but the hospital must verify and approve its applicable rule."

Q29. How is the midnight census used?

"It is an exact point-in-time patient count at 12:00 AM. We do not enter a nurse count at midnight or apply a staffing ratio there."

Q30. Does the application derive census from admissions, discharges, or transfers?

"No. Each handover census is entered as observed. We did not assume movement records are available or implement a movement-based formula."

Q31. Does approving a suggestion move a nurse?

"No. It records human review of an aggregate possibility only. It does not select a person, check competencies, update actual counts, or change a roster."

Q32. What prevents a suggested move from putting the sending unit below its requirement?

"The service calculates surplus only above the sending unit's configured target and checks cumulative approvals again. It checks the receiving requirement, snapshot freshness, and whether the unit list has changed. A qualified human still has to verify safety."

Q33. Are Emergency visits and OPD included?

"OPD is excluded. Emergency census is for admitted or observation-bed patients; walk-in visits are not part of the inpatient count."

Q34. Why not use WISN alone to redeploy staff during a shift?

"WISN uses annual workload and available working time for longer-term planning. Shift coverage uses directly observed census and actual on-duty nurses. The project keeps those time horizons separate."

Q35. Is this multi-hospital?

"No. It is a single-facility prototype. The current user accounts share the same department data; there is no Hospital model or tenant boundary."

Q36. Are patient identities stored?

"The operational schema has aggregate counts, not patient identifier fields. It has free-text notes, so the UI warns users not to enter names or identifiers."

Q37. Are only managers allowed to configure and review?

“Only an administrator can configure staffing standards. Authenticated users can enter census and record review decisions; this is not a granular nursing-manager permission system, so stronger role controls are needed before deployment.”

Q38. Is email verification enforced?

“The routes use Laravel's `verified` middleware name, but the current User model has the `MustVerifyEmail` implementation commented out. I would not claim verification is enforced until that is checked or corrected.”

Q39. How is annual turnover used?

“Turnover is a separate longer-term workforce-supply measure. I do not see a turnover table or service in this checked-in Laravel source, so it is not currently feeding the WISN calculation or operations recommendations.”

Q40. What are the main limitations?

“The data are manually entered/demo-oriented, the model covers one facility, WISN inputs and operational ratios need local validation, there is no roster/skills integration, and role-based access needs further work.”

Q41. What would you build next?

“First validate staffing standards with clinical leadership. Then add granular role-based access, a nurse roster and competency model, import reliable patient-flow and attendance data, show data freshness, and run recommendations in shadow mode before operational use.”

Q42. How do you know the calculations are correct?

“The formula is isolated in a service and a regression test checks a known example. Before claiming clinical validation, I would run the test suite and compare representative outputs with independently checked manual calculations and expert-approved inputs.”

Q43. Can this make staffing decisions automatically?

“No. It is a decision-support prototype. A qualified human reviews any potential move, and no suggestion should override patient-safety policy or professional judgment.”

17. Short presentation script

Opening (about 45-60 seconds)

“Our project is WISN Core, a Laravel-based web prototype for nursing workforce planning. It implements selected WHO WISN calculations using department working-time components, annual workload volumes, and activity time standards. The system calculates staff requirements and a WISN ratio, then presents results in a dashboard and a PDF report. Operations is a separate manual inpatient census and shift-staffing view, with counts at midnight and the 7 AM and 7 PM handovers. Required nurses come only from a locally approved, effective target; aggregate move suggestions require human review. It does not connect to a live hospital system or assign individual nurses, so our focus is an academic prototype and a foundation for future validation and integration.”

Formula explanation (about 30 seconds)

“First, we calculate annual available working time for one nurse. For each health-service activity, we divide AWT by the time standard to get the standard workload, then divide annual volume by that standard workload to get required FTE. We adjust for support time with CAF, add the code's additional-activity FTE, and compare the total with current staff using the WISN ratio.”

Closing (about 20 seconds)

“The key contribution is making these inputs and calculations visible and repeatable through a browser-based tool. The next step is validating local standards and connecting trustworthy roster and patient-flow data while keeping clinical approval and safety rules in control.”

18. Last-minute rehearsal checklist

Explain

- ☐ ☐ I can say what WISN Core does in one minute.
- ☐ ☐ I know the difference between WISN and the manual operations overlay.
- ☐ ☐ I can define AWT, time standard, annual volume, FTE, CAF, AAF, and WISN ratio.
- ☐ ☐ I can calculate the 1,696-hour AWT example.
- ☐ ☐ I can explain why the ratio below 1 means the entered current headcount is below the estimated requirement.

Show

- ☐ ☐ I know how to log in and open the correct screens.
- ☐ ☐ The demo database is prepared and contains clearly labelled sample data.
- ☐ ☐ I can show a department, its activities, dashboard result, and PDF report.
- ☐ ☐ I can demonstrate Operations with hypothetical data and state that it is not a live feed.
- ☐ ☐ I have a backup screenshot/PDF if the network or local database fails.

Defend honestly

- ☐ ☐ I will call seed values simulated/demo values unless I can verify their source.
- ☐ ☐ I will not claim an EHR integration, automatic deployment, or clinical validation that is not in the code.
- ☐ ☐ I know all users currently share department data and there is no manager-role authorization.
- ☐ ☐ I will not enter or show patient names/identifiers.
- ☐ ☐ I ran `php artisan test` on a working environment—or will say clearly if I could not run it.
- ☐ ☐ I have checked the email-verification caveat before claiming verified accounts.

19. Quick glossary

Word	Meaning in this project
AWT	Available Working Time; annual hours available per nurse after deducted leave/non-working time.
Blade	Laravel's server-side HTML template language.

Word	Meaning in this project
CAF	Category Allowance Factor; in this code, multiplier for the sum of support-time fractions.
Controller	Class that receives a web request, validates/orchestrates it, and returns a response.
Eloquent	Laravel's ORM for reading/writing related database records.
FTE	Full-time equivalent, a workload-sized staffing unit.
Migration	Versioned script that creates or changes database tables.
Middleware	Request filter such as authentication.
Model	PHP class representing a database entity/table.
ORM	Object-relational mapper: links code objects to relational database rows.
Route	URL-to-controller mapping.
Seeder	Script that inserts example or initial database rows.
Service	Class that holds application/business logic, such as the WISN calculation.
Time standard	Entered time used for an activity in the staffing calculation.
WISN ratio	Current staff divided by the project's calculated requirement.

20. Final sources to keep open during teaching

- [ARCHITECTURE.md](#) — implementation architecture and the new operational overlay.
- [WISN_Core_Project_Report.md](#) — project rationale, objectives, methodology context, and design discussion.
- [WISN_Proposal_Final \(1\).pdf](#) — proposal and academic framing.
- [9789240070066-eng.pdf](#) — WHO WISN reference included in the repository.
- [Nepal Health Facility Data Summary.pdf](#) and [WISN_Core_Source_Access_Sheet.pdf](#) — Nepal context/source material bundled with the project.
- [composer.json](#) and [package.json](#) — authoritative dependency manifests.
- [routes/web.php](#), [app/Services/WisnCalculatorService.php](#), models, migrations, controllers, and views — authoritative current implementation.

Final coaching reminder: confident presenters do not claim that a prototype does more than it does. Explain the method, show the code path, label the sample data, and describe the limitations as planned engineering work.